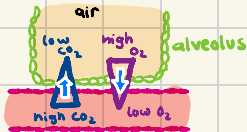


gas exchange

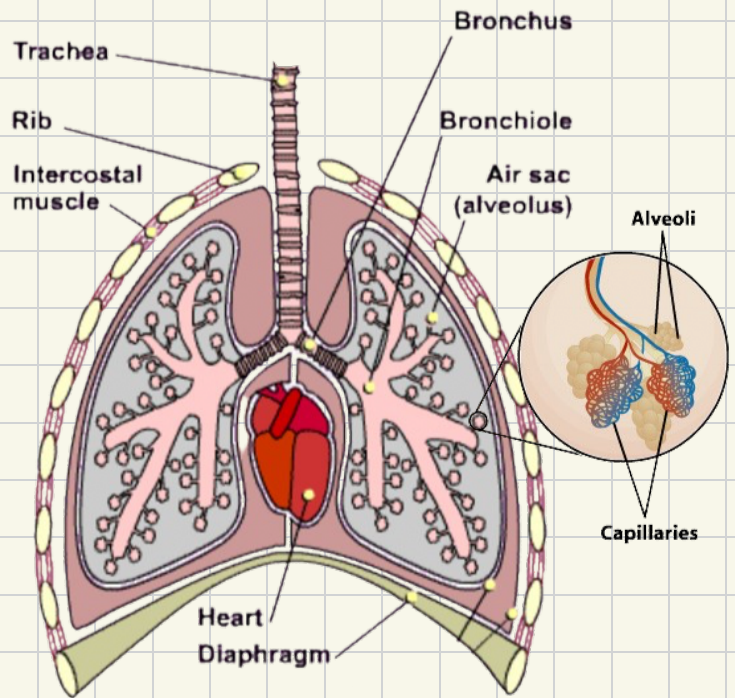
Function of the gas exchange system:

- i) filtering the dust & pathogens in the air
if they enter the lungs, the alveoli can get damaged. These dust & pathogens are filtered out as they pass the trachea
- ii) minimize the distance of diffusion between air & blood

* gas exchange must happen all the time!

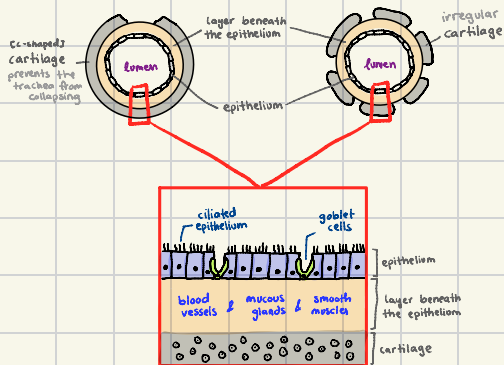


- iii) maintain the gradient between air in alveoli & the blood in capillaries



Trachea

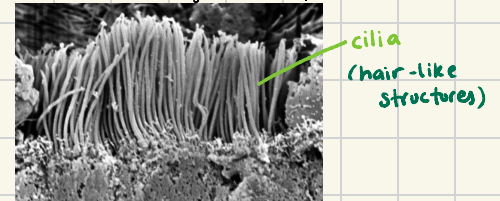
Bronchus



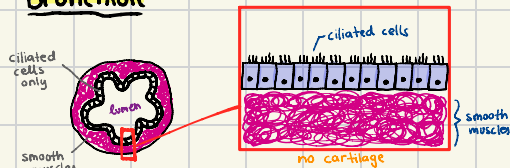
light microscope



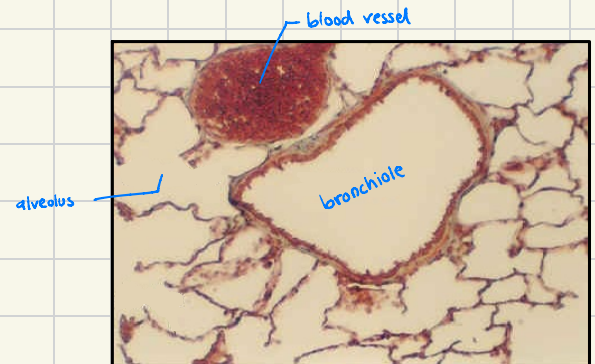
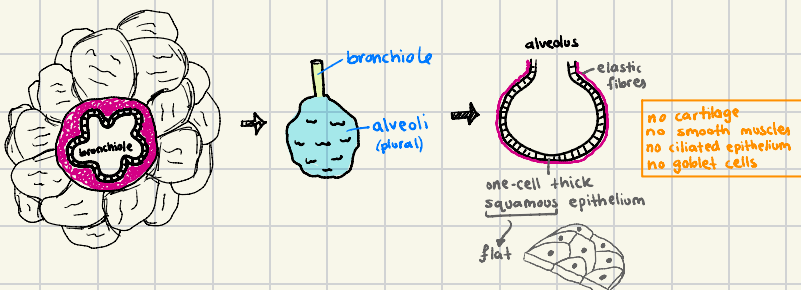
SEM (scanning electron microscope)



Bronchiole

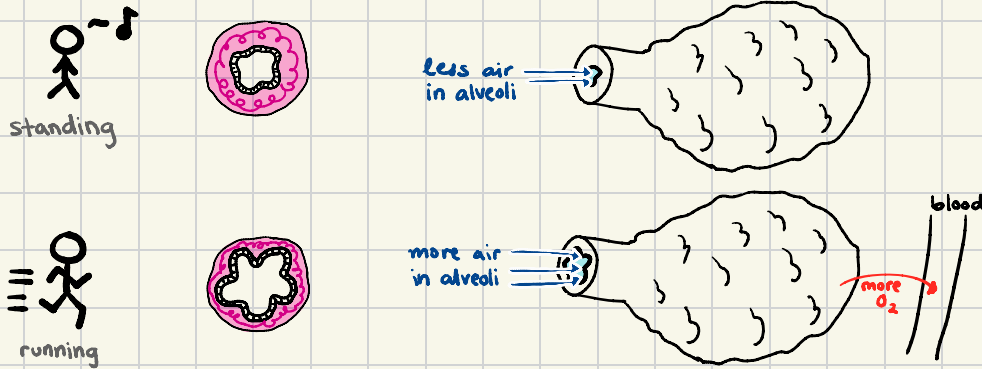


Alveoli

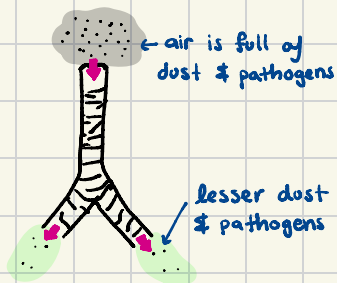


Summary	trachea	bronchus	bronchioles	alveoli
epithelium	- ciliated epithelium - goblet cells		- ciliated epithelium	- squamous epithelium
beneath epithelium	blood vessels & mucous glands & smooth muscles		blood vessels smooth muscles	elastic fibres
cartilage	c-shaped	irregular	X	X

Response of bronchioles during exercise

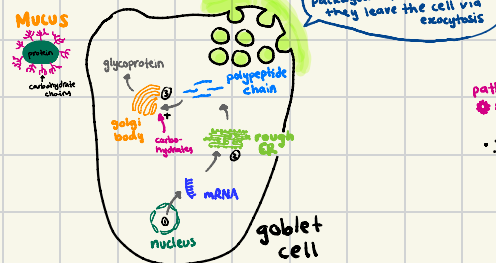


Filtering of dust & pathogens

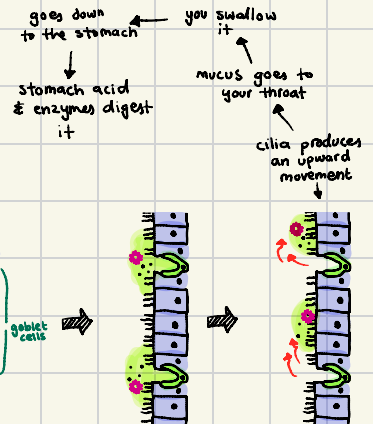


How?

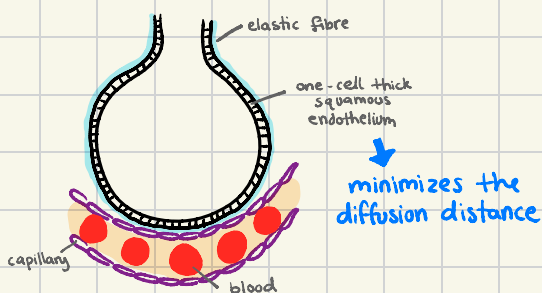
⇒ goblet cells



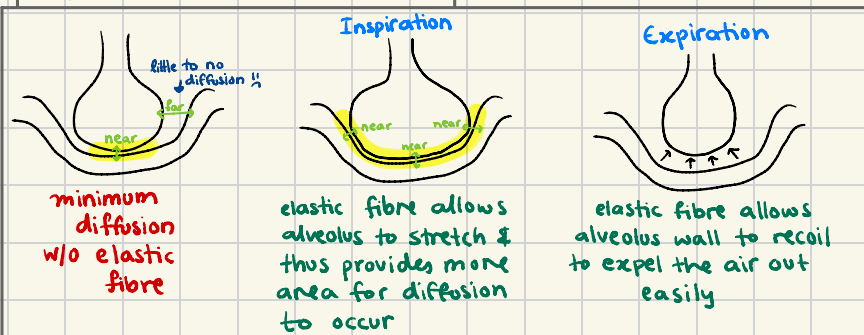
too much dust & pathogens will trigger the brain to sneeze/cough to get it out through your mouth



Minimizing diffusion distance

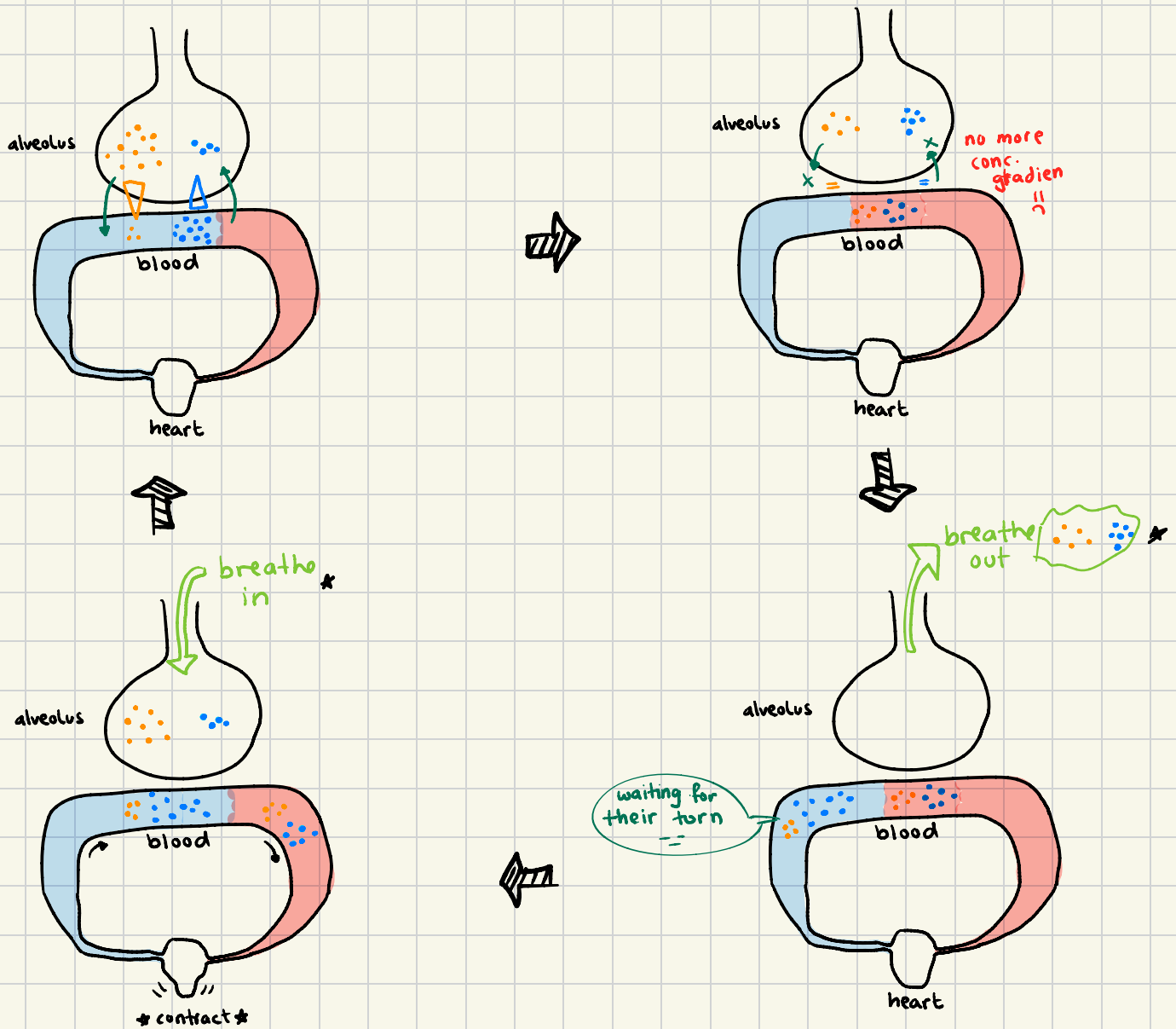


why is the elastic fibre important?



Maintaining the conc. gradient

oxygen
CO₂



★ To maintain the conc. gradient:

- ① lung ventilation ensures that the alveolus will always have a high PP_{O_2} & low PP_{CO_2}
- ② heart contraction pushes deoxygenated blood to lungs, ensuring that blood has a low PP_{O_2} & a high PP_{CO_2}